

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech Examination (EE)

March-April 2018

EE 301 Electrical Machines-II

Date: 27.03.2018, Tuesday Time: 1:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1** Draw and explain the torque speed characteristics of induction motor. [05]
- Q – 2 (a)** Explain the working of star delta starter used for starting the polyphase induction motor [05]
- Q – 2 (b)** Explain the cascade connection method used for speed control of polyphase induction motor [05]
- Q – 2 (c)** Define the slip. Explain the methods for measurement of slip of induction motor. [05]

OR

- Q – 2** Draw the circle diagram for a 3 phase, 6 pole, 50 Hz, 400V, star connected induction motor from the following data (line values): No load test : 400V; 9 A, 1250 Watts. Short circuit test : 200 V, 50 A, 6930 Watts. The stator copper loss at standstill is 55 % of total copper losses and full load current is 32 A. From the circle diagram determine (i) power factor, slip, output, efficiency, speed and torque at full load (ii) starting torque (iii) maximum power factor (iv) maximum power output (v) maximum power input (vi) maximum torque in synchronous watts and slip for maximum torque. [15]
- Q – 3** Attempt any Three (Each carry five marks) [15]
- (i) Write a note on induction generator.
 - (ii) Explain the cogging and crawling.
 - (iii) Explain the power stages of induction motor.
 - (iv) Write the torque equation of induction motor under running condition. Derive the condition for maximum torque at starting and during running condition.

SECTION – II

- Q – 4** Draw and explain the equivalent circuit of single phase induction motor based on double revolving field theory. **[05]**
- Q – 5 (a)** Explain the working and construction of shaded pole motor. **[05]**
- Q – 5 (b)** Write a note on double cage induction motor. **[05]**
- OR**
- Q – 5 (b)** Write a note on universal motor. **[05]**
- Q – 5 (c)** Write a note on linear induction motor. **[05]**
- OR**
- Q – 5 (c)** Give comparison between squirrel cage and slip ring induction motor. **[05]**
- Q – 6** **Attempt any Three (Each carry five marks)** **[15]**
- (i)** Explain the construction and working of Schrage motor.
 - (ii)** Explain the construction, working and application of permanent capacitor induction motor. Draw the phasor diagram.
 - (iii)** Write a note on stepper motor.
 - (iv)** Write a note on repulsion motor.
